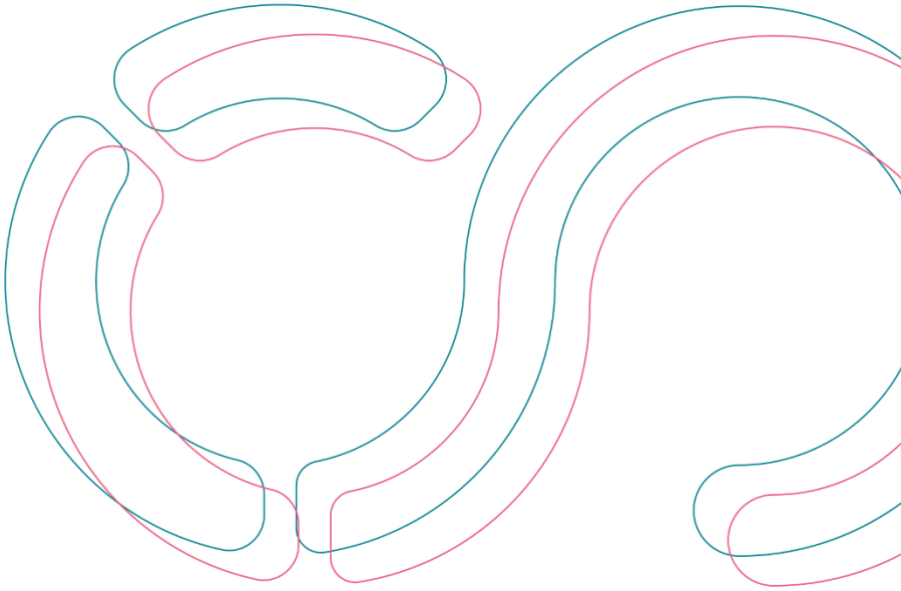
eosc | GRAVITY

Cascading grant project: Preparatory grant

Project acronym: EOSC-ES-Prep

Project deliverable: Project Charter



Funded by the European Union

Grant agreement ID: 101188045

Cascading project name and acronym	EOSC-ES-Prep
Grant ID of the cascading project	EOSCGRAVITY-PREP-2026-010-EOSC-ES-PREP
Lead beneficiary	IACS (Instituto Aragonés de Ciencias de la Salud)
Authors	Juan González-García (IACS), Ignacio Blanquer-Espert (UPV), Daniel Garijo (UPM), Helena Lozano (CSIC), Gorka Epelde-Unanue (Biogipuzkoa) on behalf of the EOSC-ES-Prep consortium
Due date	M5 - September 2026
Approved on	

Disclaimer

This project has been funded within the Cascading Grants under the EOSC Gravity Project, which has received funding from the European Union's Horizon Europe programme under Grant Agreement No. 101188045.

The views and opinions expressed herein are those of the author(s) only and do not necessarily reflect those of the European Union, the granting authority and the EOSC Gravity partners. Neither the European Union, nor the granting authority or EOSC Gravity partners can be held responsible for them.

The deliverables are assessed solely in accordance with the formal criteria of the call. The review process seeks to maintain an impartial and unbiased position and to avoid any conflict of interest. It does not include the evaluation or endorsement of the technical or operational solutions proposed by the grantees. Consequently, the reviewers cannot be held responsible for any inaccuracies, shortcomings, or errors that may appear in the submitted materials or related project documentation.

Copyright notice

Except where otherwise stated, this work is licensed under CC BY 4.0.

1 Project summary

The EOSC-ES Node aims to be the Spanish national Node of the EOSC Federation. Its purpose is to turn a large but fragmented national Open Science ecosystem into a single, governed and interoperable entry point through which Spanish FAIR data, research software, repositories and analysis services become discoverable and reusable across the Federation, and Spanish researchers reach the resources of other Nodes.

The scope is national and deliberately multi-domain: the Node federates existing national assets rather than creating new centralised infrastructure. It draws on an ecosystem including unique scientific and technical infrastructures (ICTS) and the Red Española de Supercomputación (RES); the national research and education network RedIRIS (Red.es) and its SIR federated identity service; national nodes of European Research Infrastructure Consortia (ELIXIR, BBMRI, ECRIN, EATRIS, CLARIN and DARIAH through CLARIAH-ES); and national thematic data spaces in health, marine sciences, environment and social sciences and humanities. A list of the initial resources onboarded at enrolment are listed in Section 3.

The Node will implement the mandatory Federating Capabilities FC-1 (AAI) and FC-2 (Resource Catalogues and Registry services), and prepare FC-3 (Helpdesk), FC-4 (Monitoring) and FC-5 (Service Management) ahead of their transition to mandatory status. Node Exchange contributions comprise domain repositories and data spaces, FAIR assessment and semantic tooling, secure processing environments, AI platforms, HPC and storage capacity, and training and competence-centre functions.

EOSC-ES will contribute to the Federation a nationally coordinated multi-domain user base, a "federation within the Federation" governance model designed for a distributed research ecosystem, and an established bridge to the Spanish-speaking research community. EOSC-ES implements the Spanish Open Science Strategy (ENCA 2023–2027) and the State Plan PEICTI 2024–2027, both of which prioritise integration into EOSC. The representing entity, which will sign the EOSC Federation MoU on behalf of the participating institutions, will be designated through the process that opens at the national tripartite event of 2 October 2026; IACS acts as interim coordinator.

Value proposition

Community organisation. Spain's EOSC engagement is extensive but dispersed: 25 Spanish institutions are members or observers of the EOSC Association, nearly 40 experts have served in EOSC-A Task Forces (6 as co-chairs), and Spanish institutions lead 6 and participate in 31 of the 44 EC-funded EOSC projects (70%, CORDIS). The gap is coordination, not capacity: no institution acts as national interface to the Federation. The Node closes it: one interlocutor for governance, coordinated catalogue endpoints, one AAI integration path, one helpdesk entry, behind which 22 founding institutions onboard progressively. Eight are not yet EOSC-A members or observers, and membership is not required to join, so EOSC-ES also feeds new organisations into the Association.

Bridge to Latin America. Spanish has 635 million speakers, mostly outside Europe (Instituto Cervantes, 2025). Standard databases index ~37K Open Access Spanish publications, ~85K including regional ones. Spain joined LA Referencia —Latin America's federated repository network— in 2020 through FECYT, fully integrating its outputs. Exposing the Node catalogue —through Federating Capability 2— unifies European and Latin American aggregation along an established, governed discovery path, unique among candidate Nodes. The same route serves research in the co-official languages: CLARIAH-ES annotation pipelines, machine translation and open models make non-English texts searchable internationally, a reusable pattern for any European community publishing in an under-resourced language.

Singular resources. The Node widens the Federation's disciplinary coverage rather than duplicating it. In health, population-scale governed-access platforms under a regional secondary-use mandate (BIGAN, IACS), clinical and one-health data spaces (OHSIRIS, IBiS-HUVM; FEDERA Salud, IBiS-HUVR; BIGAN-OneHealth, IACS; EUCAIM, UPV) and the Spanish nodes of BBMRI (IMIB), EATRIS (IRYCIS), ECRIN (FIMABIS) and ELIXIR (BSC-CNS) provide real-world longitudinal data with the governance and secure-processing environments the European Health Data Space (EHDS) regulation requires. In environment and marine sciences, CSIC operates observation networks with series reaching back to the 1930s, plus data-space demonstrators for marine research (NEREIDAS, UMU) and territorial data (HELEADE, UA). In computing and physics, PIC/CIEMAT contributes CosmoHub (Gaia, Euclid, LSST, CTA) and a Rucio-based Scientific Data Lake over RES storage; BSC-CNS, CESGA and SCAYLE add HPC, cloud and AI capacity.

Beneficiaries and sustainability. Researchers, technicians and managers in Spain and across Europe gain a single route to Spanish data, services and compute; research-performing organisations and universities gain European visibility without operating their own Federation interface; public bodies and funders gain an instrument to implement and monitor ENCA and PEICTI commitments; service providers reach a wider user base through shared federating capabilities; and citizens, through data spaces addressing health, environmental and societal challenges. Sustainability rests on a defined leadership structure, transparent in-kind commitments and cost-sharing, and the progressive integration of additional public administrations, so the Node does not depend on project funding. Spain is moreover well positioned in the Common European Data Spaces the Node connects to: health, Green Deal, Languages (ALT-EDIC) and Cultural Heritage (Europeana).

2 Repositories and services delivered

For space constraints, the table lists a selection of Node Exchange repositories and services — from a total of 28— selected by their maturity and alignment with Use Cases and Federating Capabilities. Node Core functions are described in Section 5. All resources are operational assets self-assessed at TRL 7 or above. The fourth column adds, after "Users:", the direct value for researchers.

Resource ID	Resource Description	Resource Access Policies	Federation Contributions & Value to Users	TRL
ES-RES01 Compute and storage	HPC, cloud and storage of SCAYLE, BSC-CNS and CESGA for the use cases. CSIC and CESGA are part of IBERGRID.	European-wide through the open RES calls: free for open R&D, granted per activity.	Horizontal capacity for ES-UC1 to ES-UC4. Users: free compute for open R&D via one European call.	9
ES-RES02 Scientific Data Lake	Rucio data management over federated storage on RES elements (PIC/CIEMAT).	European-wide via OIDC and eduGAIN; per-dataset control; WebDAV, S3, xrootd.	FC-1, FC-2; federated namespace, PIDs, provenance. Shares the ESCAPE backbone: bridge to the CERN Node (ES-UC3). Users: share and cite large data with own identity.	7
ES-RES03 FAIR suite	FOOPS!, RSFC and FAIROs: FAIRness of vocabularies, code (FAIR4RS) and Research Objects (UPM).	Open, European-wide; GitHub action; in RO-Hub (PSNC).	FC-2. FAIR quality gate for the Node catalogue (ES-D1); in process with the Poland Node. Users: FAIRness feedback before publishing.	8–9
ES-RES04 RAISE	Code runs in a Trusted Execution Environment at the data holder; only audited results leave (Biogipuzkoa).	European-wide; open with traceability, or owner-approved revocable access.	Script-shipping over sensitive data, only partially covered by the Federation today. Serves ES-UC1. Users: analyse sensitive data that cannot leave the holder.	8–9
ES-RES05 AI4EOSC	Open ecosystem for deploying and accessing research-grade AI services across EOSC (CSIC).	European-wide, eduGAIN with quota; community gateways.	FC-2 and the Node's candidate FC-6 demonstrator. Supports ES-UC1 / ES-UC2. Users: AI models without running their own infrastructure.	9
ES-RES06 Privacy toolbox	Anonymisation tools for tabular data: Anjana, pyCANON, trasgoDP (CSIC).	Open, European-wide; open source.	Privacy gate for governed-access data; direct enabler of ES-UC1. Users: anonymise tabular data before sharing it.	9
ES-RES07 CosmoHub	Exploration of large astronomical datasets: 200+ TB, 300+ billion objects (PIC/CIEMAT).	European-wide via eduGAIN; OIDC per-catalogue control; open and controlled releases.	FC-1, FC-2; IVOA standards. Gaia, Euclid, LSST, CTA. Anchors ES-UC3. Users: query billions of objects without downloading.	8
ES-RES08 NEREIDAS	Marine sciences data space for the south-western Mediterranean (UMU).	Registered members; some use cases open.	FC-2, integrated by reference via Blue-Cloud 2026. Anchors ES-UC2. Users: Mediterranean marine data behind one access path.	7
ES-RES09 BIGAN	Population-level longitudinal health data of Aragón, with a secure processing environment (IACS).	Governed, European-wide: ethics approval and data permit; fees apply.	SPE aligned with EOSC and EHDS: analysis in situ, no extraction. FC-2. Anchors ES-UC1. Users: real-world health data in a secure environment.	8
ES-RES10 CLARIAH-ES repositories	Two CoreTrustSeal DSpace repositories of language and heritage resources; CMDI, PIDs, OAI-PMH (UPV/EHU, ULPGC).	Open by default; eduGAIN / CLARIN SPF for restricted resources.	FC-1, FC-2, FC-7. Dual visibility in the Resource Hub and in CLARIN. Anchors ES-UC4. Users: trusted, citable deposit of language resources.	8–9
ES-RES11 Annotation pipeline	REST services for lemmatisation, morphosyntax, NER and parsing in the official languages (CLARIAH-ES).	Open for moderate volumes; quotas for massive processing.	FC-2. EOSC IF composability and the cross-lingual layer of Section 2. Serves ES-UC4. Users: annotation in the official languages, no install.	7–8
ES-RES12 Training and K-centre	Spanish CLARIN K-centre (CLARIAH-ES): tutorials and course materials on language resources and their annotation (UPV/EHU, ULPGC).	Open, European-wide; materials under open licences; advice through the Node helpdesk.	Research Training and Competence Centre resources; thematic SSH second level for FC-3; materials to the EOSC Academy (ES-D5). Serves ES-UC4. Users: tutorials, course materials and expert advice.	8

3 Use cases

The Node commits to four main use cases during the build-up phase, spanning health, marine sciences, astrophysics and the social sciences and humanities. All four are inter-node, each with identified counterpart Nodes, meeting the enrolment requirement of at least one multi-Node case with margin. All four federate assets already in operation, cover the Federation's three access regimes — open, controlled under a collaboration policy, governed under statutory secondary-use rules — and follow one pattern: discovery in the Node catalogue and the EOSC Resource Hub (FC-2), authentication through the Node's federated identity (FC-1), analysis of the data where they reside, and deposit citable FAIR Research Products as FAIR Digital Objects (FDOs).

Three of them connect EOSC —the transverse data space for research and innovation— to sectoral Common European Data Spaces: ES-UC1 to the European Health Data Space, ES-UC2 to the marine component of the Green Deal data space, ES-UC4 to the language and cultural heritage data spaces. ES-UC3 has none behind it. This answers the duplication risk registered in Section 6: a research layer over the sectoral data spaces, not a copy. Cases not selected (Geo, Plant, Animal, Ecosystem Health; other CLARIAH-ES scenarios) stay in the pipeline.

Use Case ID	Use Case Description	Federation Contributions & Value to Users	Participating organisations	Other Nodes involved
ES-UC1 EHDS: secondary use of health data, One Health scope	Federated pathway for the secondary use of health data, aligning EOSC resources with the EHDS Regulation, whose own catalogue for secondary use already spans One Health: environmental and behavioural determinants of health and data on pathogens impacting human health, not only clinical records. It builds on the EHDS pilot prototyped in EOSC United, on the EOSC-A Health Data Task Force propositions, and on EUCAIM, a working precedent of federated catalogue, query, access negotiation and processing over governed-access data, with Spanish participation (UPV). Two levels. Intra-node: through the Spanish national nodes of the health research infrastructures, alignment of the regional health data spaces and their secure processing environments with the emerging national health data space (the Spanish EHDS node), and with the One Health data space at CESGA, which pools human, animal and environmental health data under one access path. Inter-node: the national health data spaces of other Member States through HealthData@EU, and the Nodes holding life-science data and distributed analysis capacity. Discovery through (Health)DCAT-AP; analysis	Links the Federation to a regulated Common European Data Space with statutory governance and its own infrastructure programme, positioning EOSC as its research-facing discovery and analysis layer. It composes with the Life Sciences workflow environment (Galaxy, tested in the EOSC United EHDS pilot) and the BBMRI distributed-analysis pattern (MCVAL). The One Health scope is what a Node can demonstrate and a thematic infrastructure cannot: human, animal and environmental health data sit under different regimes and custodians, and the Node puts them behind one identity, one catalogue and one analysis pattern, with 1HealthAI as compute anchor and ES-UC2 as its marine counterpart. Value to users: wider data, tools, secure environments and publication routes for health research, and cross-border, cross-domain	IACS (lead) — EOSC United / CDMb, ASPIRE, in-house SPE / BIGAN; CSIC — EOSC-SIESTA; BSC-CNS — EOSC ENTRUST, BSC AI Factory; CESGA — 1HealthAI AI Factory; UPV — EUCAIM; Biogipuzkoa - RAISE; regional health data spaces: BIGAN / BIGAN-OneHealth (IACS), OHSIRIS (HUVM-IBiS), FEDERA Salud (HUVR-IBiS), HealthData@MAD-R&I (IRYCIS); regional environmental data spaces: HELEADE (UA).	Life Sciences Connect (EOSC ENTRUST); BBMRI-ERIC; EOSC EU Node (orchestrating)

	where the data reside, so they never leave the holder.	studies without moving the data. Common European Data Space: European Health Data Space.		
ES-UC2 Ocean Health	Intelligent, integrated and scalable platform that collects and analyses multidisciplinary ocean data — physical, chemical, biological and geological — and integrates them into operational decision-support systems, applying AI to heterogeneous databases and existing monitoring tools. It builds on an observation network with time series reaching back to the 1930s and on NEREIDAS, the national marine-science data space demonstrator in the south-western Mediterranean, already in contact with Blue-Cloud 2026.	Contributes the integration of heterogeneous, long-term observational sources under a single discovery and access path, and the reuse of a national data space as a Node resource. Value to users: patterns of ocean variability, anticipation of ecosystem change and early warnings for extreme marine events, with interoperability towards European observation networks (EMODnet, Copernicus Marine, Euro-Argo and EMSO ERICs, SeaDataNet). Common European Data Space: Green Deal, marine component.	CSIC (lead) UMU — NEREIDAS	European Digital Twin of the Ocean Node (first wave) Blue-Cloud 2026 / CNR as the EOSC service platform already engaged with NEREIDAS
ES-UC3 Euclid DR1	Support to a task force of the Euclid Consortium investigating data analysis issues as part of the preparation of Data Release 1, whose membership is dynamic and controlled by the task force itself — a pattern federated infrastructure has to accommodate. The Node provides storage with federated AuthN/AuthZ through eduGAIN and OIDC, elastic quota (1 TB to 1000 TB), access over HTTPS/WebDAV, and analysis next to the data: JupyterHub for interactive work, CosmoHub for massive tabular exploration.	Demonstrates at scale compute-near-data and elastic, policy-driven provisioning of federated storage over a Rucio backbone implementing the ESCAPE Data Lake architecture, shortening the integration path with the CERN Node; exercises FC-1 with a real collaboration policy (controlled-access survey data and task force membership management) and FC-2 through IVOA-compliant catalogue exposure. Value to users: a distributed collaboration obtains in days a federated shared workspace, with reproducible and citable results. Common European Data Space: none sectoral — EOSC as the transverse data space for research and innovation.	PIC — Port d'Informació Científica (lead) CIEMAT	CERN Node — through the ESCAPE Data Lake and the shared Rucio backbone

ES-UC4 Parliamentary corpora	Creation, enrichment and federated analysis of corpora of Spanish parliamentary proceedings — Congreso and regional chambers, including the co-official languages — following the TEI schemes of ParlaMint, the CLARIN flagship for comparable parliamentary corpora. Corpora are discovered in the Node catalogue and the CLARIN VLO, processed with the annotation pipelines and the open LLM inference service for the official languages, and deposited as Research Products with PIDs in the Node's CLARIN repositories.	The corpus joins a family of comparable resources maintained by more than twenty countries — federating at Node level what is already federated at thematic level. Exercises FC-1, FC-2 and, through the DSpace repositories, FC-7, and demonstrates the composability pursued by the EOSC Interoperability Framework (repository to tool to new Research Product) in SSH, under-represented in the Federation. Value to users: comparative study of political discourse, terminological evolution and the use of the co-official languages. Common European Data Spaces: language data space (ALT-EDIC) and cultural heritage data space (Europeana Foundation).	HiTZ (UPV/EHU) (lead), on behalf of CLARIAH-ES IATEXT (ULPGC) CLARIAH-ES centres with experience in parliamentary corpora	EOSC Node Netherlands (SURF; ParlaMint use case with CLARIN ERIC); EOSC Node Slovenia (CLARIN.SI, ParlaMint repository); CLARIN and DARIAH ERICs via CLARIAH-ES

3.1 In scope

The four use cases delivered end to end during the build-up phase —see Section 9—: two by month twelve, the other two by month eighteen, reported together in ES-D4 on the capabilities exercised, the resources combined and the lessons for the Federation. For each: exposure of its resources through the Node catalogue with access policies and TRL, integration with the Node AAI, the metadata profiles for cross-domain discovery (HealthDCAT-AP, IVOA, CMDI), coordination with the counterpart Node, secure analysis close to the data with the privacy tooling required to operate it lawfully, publication of the results as FAIR Research Products with PIDs, and support to the communities behind each case. All resources involved are at TRL 7 or above.

3.2 Out of scope

The Node does not implement the statutory infrastructure of the EHDS nor act as a Health Data Access Body, and gives no ethics or legal approval for data reuse: designation, governance and permits lie with the competent authorities and each data provider. Nor does it operate observation infrastructure, curate the underlying data, produce the scientific results of the missions it supports, train foundation models, or fork the software it operates. Resources below TRL 7 and Federating Capabilities where the Node only participates stay out.

4 Compliance with technical requirements

The Node meets the minimum requirements for enrolment through a deliberate strategy of reuse: every mandatory capability is mapped onto an existing, operational national system rather than onto a new deployment. This keeps build-up cost and risk low, shortens time to enrolment, and makes sustainability a question of maintaining services that already have owners and budgets.

Enrolment criteria. As a national candidate Node from a Member State, EOSC-ES addresses the enrolment criteria as follows.

Capacity to act as a community gateway is provided by the national coordination structure described below and by the resource portfolio in Section 3. Technical specifications for AAI and catalogues are met through FC-1 and FC-2 as set out in this section. All resources onboarded at enrolment are at TRL 7 or above. Individuals for the roles of coordinator, operations, legal, cybersecurity and communications officer will be selected among the participating institutions and named before the enrolment application (Section 10). Four multi-Node use cases committed (Section 4), with intra-node alignment (ES-UC1) and a pipeline of 6 use cases. The designated representing entity commits to signing the Federation MoU. Endorsement by the competent national authority will be secured ahead of the enrolment application through the proactive involvement of the Director General for the Internationalization of Science and Innovation of the Ministry of Science, Innovation and Universities (also Spanish representative of the EOSC Steering Board) and the Spanish Commission on Open Science (COS-Gob), responsible body for the implementation of the Spanish Strategy of Open Science, both of which have already supported the creation of the Spanish Node and its principles.

Governance. The EOSC-ES Node will be represented by a designated legal entity that will sign the EOSC Federation MoU and accept the Rules of Participation on behalf of the participating institutions. The model will be a lightweight, open collaboration agreement with a general assembly, a strategic committee — the natural place for funding bodies — and a technical committee; a single director provided by the representing organisation and a chair of the general assembly. A dedicated legal entity remains under study for a later stage. The agreement is designed as a framework in which new institutions can easily support joint participation as third parties in projects. A draft of this agreement, already agreed by the participants, will be available by the submission of the Node proposal.

Node Core Capabilities. The Node operates a user entry point exposing the national catalogue, user and project management built on federated identity, and usage reporting. Horizontal compute and storage are contributed by RES and IBERGRID federated computing members rather than operated centrally by the Node; persistent identifier services are sourced from existing national and international providers. A helpdesk and a Monitoring service are defined as part of the node, following the EOSC Federation architecture. Service Management Systems (SMS) will comprise the existing SMS of the production services, along with the ones from the new services. Federating Capabilities. FC-1 and FC-2 are mandatory; FC-3, FC-4 and FC-5 are currently recommended and become mandatory in 2026, and are therefore planned as mandatory.

FC	Capability	Implementation
FC-1	AAI (mandatory)	The EOSC-ES Node will build on RedIRIS SIR, an operational identity federation integrated with eduGAIN and based on an architecture consistent with federated AAI principles. Several candidate resources already use eduGAIN, reducing the integration effort for a significant part of the initial service portfolio. The integration architecture between SIR, the EOSC Federated AAI and EOSC-ES services, including protocols, attributes, authorisation and proxy functions, will be validated during the build-up phase. Any additional component or operational responsibility beyond the current SIR service will require prior technical and institutional agreement.
FC-2	Resource Catalogues and Registry services (mandatory)	The EOSC-ES Node will reuse existing national cataloguing and harvesting capabilities, including RECOLECTA, datos.gob.es (the national DCAT-AP open-data catalogue), PLATICA and other relevant national or thematic data registries, as the starting point for FC-2. Several of these registries are already harvested by OpenAIRE, so their records already reach the EOSC EU Node catalogue; the Node will build on that path rather than duplicate it. Participating resource owners will be encouraged to expose their metadata through agreed interoperable profiles and established aggregation channels, including OpenAIRE and the EOSC Resource Hub where applicable. PLATICA (concretely CARREDI, its data repository) will be progressively integrated as its catalogue and repository components reach production readiness. During the build-up phase, the Node will define the minimum metadata profile, validation process, harvesting model and allocation of responsibilities required for interoperability with the EOSC Federation catalogues. Integration of the future Spanish EHDS metadata registry will be assessed in coordination with the competent organisations.
FC-3	Helpdesk (mandatory in 2026)	A Zammad deployment operated by UPV/IACS/UJAEN, interoperable with the helpdesks of individual services and integrated with the federated helpdesk through inter-node ticket exchange. Second-level support provided by resource owners, with the Spanish CLARIN K-centre as thematic SSH support. A support group per service will be defined by the responsible institution finally selected.
FC-4	Service Monitoring (mandatory in 2026)	ARGO Monitoring Framework with Prometheus/OpenMetrics exporters, integrated with the EOSC Service Registry so that services are mapped to probes of the EOSC-ES services. Availability of compatible probes is a condition of onboarding.
FC-5	Service Management System (mandatory in 2026)	FitSM-based SMS with documented procedures, service profiles with PIDs and a peer-review onboarding process, including security coordination between Nodes. The existing production-ready services already implement SMS guidelines that will be compiled and revised with respect to the EOSC-EU node guidelines.
FC-6	Service Accounting (recommended)	Candidate for demonstration through the AI and inference platforms, where consumption in GPU-hours or tokens is already metered and a credit model is under evaluation. Adoption planned during the build-up phase.
FC-7	Research Product Accounting (recommended)	DSpace-based repositories already produce standardisable usage statistics, as do national repositories aggregated through RECOLECTA. Adoption planned during the build-up phase.
FC-8 / FC-9	Order Management / Application Deployment (recommended)	Not committed at enrolment. UPV has developed and operates the Application Workload Manager (AWM) service of the EOSC EU Node, which instantiates resources through the tool hub of the EOSC EU Node, which is directly relevant to FC-9. It also operates as AWM in the IBERGRID infrastructure and it can be used with any other computing resource of the federation if it uses any of the supported backends (https://www.grycap.upv.es/im/index.php)

5 External dependencies & key risks

External Dependencies & Risks	Actions / mitigation measures	Deadline
The technical integration model between RedIRIS SIR, eduGAIN, the EOSC EU Node AAI and the services onboarded through EOSC-ES has not yet been validated (protocols, attributes, authorisation, proxy functions).	Validate the FC-1 integration architecture jointly with the EOSC AAI stakeholders, documenting authentication flows, attribute and authorisation requirements, technical interfaces and allocation of responsibilities. Any component or operational responsibility beyond the existing SIR service requires a separate technical, organisational and resource agreement.	M1 scope; M2 architecture and roadmap
Selection of the Node catalogue and of the minimum metadata profile. Until it is fixed, resource owners cannot prepare their metadata.	Publish the minimum DCAT-AP profile and a validator; run a pilot harvest with three volunteer repositories before opening onboarding.	M3
Dependence on capabilities the Node does not provide itself: horizontal compute, storage and PIDs, and the EOSC EU Node for catalogue registration and AAI.	Map what the EOSC EU Node provides against what must be sourced nationally from RES members and formalise those commitments; early bilateral contact with the counterpart Nodes identified in Section 4, escalating through Federation governance where timelines are at risk.	M3 onwards
Low motivation to participate where resources are already aggregated elsewhere, and perceived duplication with the Common European Data Spaces.	Position the Node as a national showcase for Europe and as the cross-disciplinary discovery layer over the sectoral data spaces, not another repository; document concrete benefits per provider type in the onboarding material and scout interoperability pathways with those data spaces.	Entire project
Endorsement by the competent national authority, a mandatory criterion for national Nodes, is not secured in time for the enrolment application.	Identify the endorsing body and the procedure early and brief it on Node scope and governance; align the request with the third-wave enrolment window.	Before enrolment
Insufficient political or institutional support, or the Node perceived as belonging to a single organisation.	Early publication of the governance model and onboarding criteria; guaranteed voice for all organisations in Node governance; structured dialogue with national bodies aligned with ENCA.	M3 (ES-D1)
Coordination overhead and reliance on volunteer effort, which is viable at the start but not as an operating model.	Stabilise a working team with named roles and declared effort; simple governance, focused meetings, written decision records.	Entire project
Incomplete or unmaintained resource fiches blocking onboarding.	No named owner, no onboarding. Annual review of fiches; resources without a responsible person are withdrawn from the catalogue.	M6, annually
Rapid evolution of EOSC policies and technical guidelines, with misalignment against the minimum Node requirements discovered late.	Continuous monitoring of the Federation Handbook and Tripartite updates, with revision checkpoints at each milestone; self-assessment against the Handbook checklist at entry time (M1) and along the development (ES-MS4 at M12; ES-D5 at M24).	M1, M12, M24; continuous
Sustainability of Node core services beyond the current funding horizon.	Structured engagement with national funding agencies; alignment with the relevant Horizon Europe EOSC calls; progressive integration of additional public administrations into the cost-sharing model.	Entire project
Divergence between the EOSC Federating Capabilities and the Simpl-based stack adopted by the EHDS and other Common European Data Spaces, leaving health resources with two integration paths.	Expertise of participating institutions as the Node's Simpl test bed: map FC-1 and FC-2 onto Simpl AAI, catalogue and connectors during ES-UC1 and report it to the Federation (ES-D4); track Simpl-Open releases and the EHDS technical specifications.	M9 mapping (ES-D3); M18 report (ES-D4)
Data protection and security compliance of governed-access health resources (GDPR, EHDS, ENS), with roles between data holders, data users, SPE operators and access bodies still unallocated.	DPIA and ENS categorisation for each governed resource before onboarding, applying the role mapping and phase-by-phase governance checklist of the existing national projects (data holder / data user / intermediation entity / DAC); the Cybersecurity Officer validates security, provenance and audit requirements.	M6, annually

6 Contributions

The deliverables below are those of the EOSC-ES Node build-up phase. Timelines are given in months from the kick-off of the build-up phase (M1). Deliverables are documents, or demonstrators accompanied by documentation; the entry into production of each capability is tracked as a milestone in Section 9. Responsibilities are assigned to the institutions within the Node that contribute the corresponding components.

Deliverable ID	Deliverable Name	Responsible	Deadline
ES-D1	Node governance and third-party contribution policy (report): signed collaboration agreement; terms of reference of the general assembly, strategic committee and technical committee; designation of the legal entity that signs the EOSC Federation MoU; resource onboarding framework and third-party contribution policy — eligibility criteria (TRL 7 or above, named owner, FAIR metadata), minimum metadata profile and validator, acceptance procedure, rights and obligations of contributing providers.	[TBD: representing entity], with UPM and RedIRIS for the onboarding framework	M3
ES-D2	Use case implementation plans (report): for each of the four use cases, the scientific question, user communities, workflow across Node and counterpart resources, Federating Capabilities exercised, metadata profile, success criteria and the written agreement with the counterpart Node.	Use case lead institutions (IACS, CSIC, PIC, UPV/EHU), with the counterpart Nodes	M3
ES-D3	Node technical and operations documentation (report): integration architecture between RedIRIS SIR and the EOSC Federated AAI, metadata profile and harvesting model of the Node catalogue, with the conformance evidence of FC-1 and FC-2; helpdesk support model and inter-node ticket exchange, monitoring probe specification, FitSM-based service management procedures, security coordination procedures and data protection plan (FC-3 to FC-5); mapping of FC-1 and FC-2 onto the Simpl stack.	RedIRIS (Red.es) and FECYT (FC-1, FC-2); UPV/IACS/UJAEN and [TBD: SMS operator] (FC-3 to FC-5)	M9
ES-D4	Use case report: results of the four use cases demonstrated end to end, counterpart integration performed, capabilities exercised, resources combined and lessons for the Federation.	Use case lead institutions, with the counterpart Nodes	M18
ES-D5	Node self-assessment and updated Project Charter: assessment against the EOSC Federation minimum requirements, peer-reviewed within the Node; report of engagement, training and third-party onboarding, with the training materials and sandboxes contributed to the EOSC Academy; Charter updated with the governance, portfolio and capability status reached.	Technical committee and [TBD: representing entity], with UPM for engagement and training	M24

Funded by the European Union
Grant agreement ID: 101188045

Quality assurance. Each deliverable undergoes internal peer review by at least two participating institutions not involved in its drafting, assessing alignment with Federation requirements, completeness and internal consistency, and practical feasibility. Consolidated comments are documented, addressed by the authors and validated by the Node coordination before release.

Funded by the European Union
Grant agreement ID: 101188045

7 Community engagement

The Node's engagement model starts from an established network of 22 institutions — EOSC Association members alongside organisations with demonstrated Open Science maturity, including the Spanish nodes of ELIXIR, BBMRI, ECRIN, EATRIS, CLARIN and DARIAH. EOSC Association membership is encouraged but not required to participate, which keeps the entry barrier low.

Lessons from the current wave of Nodes shape this approach in three ways. A national Node needs one clear operational owner even when the community is large and distributed, as in the Czech Node where a single NREN coordinates a working group of dozens of centres and hundreds of members. Nodes that grow out of an already-operating community reach production faster than those assembled administratively, as in the European Digital Twin of the Ocean — which argues for anchoring the first use cases in communities that already collaborate. And Nodes are best built from a small set of services that are adapted and integrated first, with the Node emerging from them, rather than from an attempt to onboard everything at once. EOSC-ES adopts this incremental logic explicitly, in resource onboarding and in use case selection.

Engagement operates through a unified national EOSC platform acting as the hub for services, research products and community activity, interoperating with Federation catalogues and implementing EOSC AAI for single sign-on; an open onboarding route for third-party Spanish resources with published criteria and a documented procedure; dissemination through the participating institutions' own channels, operational data spaces, repositories, training environments and mailing lists; communication and training tailored to distinct profiles — research, industry, government and society — rather than a single generic message; involvement of the user community in service design from the outset and its unified representation in Node governance; and structured dialogue with national authorities and representative bodies to secure institutional endorsement without creating operational dependencies.

The participating institutions also act as multipliers through the European projects they already run. Spanish partners in EOSC and Open Science projects will incorporate the promotion of the EOSC-ES Node into the face-to-face meetings they host in Spain, within the sessions open to the public, and into their participation in Open Science events. This turns an existing calendar of national and European gatherings into recurring Node outreach at no additional cost, and reaches communities that are not yet engaged with the Federation.

In-person engagement is anchored in an annual national EOSC-ES event, with the RES Users' Conference — which attracts over 1,000 researchers — as a candidate co-location venue, complemented by onboarding workshops and cross-institutional technical meetings. At Federation level, the Node participates in cross-node coordination and thematic working groups, contributing to interoperability standards, onboarding practices and operational convergence.

8 Timing and milestones

Milestone ID	Milestone Description	Target Delivery Date
ES-MS0	Enrolment application submitted in the third wave, with the endorsement of the competent national authority and the five committed roles named.	M-3 (before M1)
ES-MS1	Node constituted and open to the initial wave of institutions: collaboration agreement signed, governance bodies constituted and onboarding framework published (ES-D1); use case implementation plans agreed with the counterpart Nodes (ES-D2).	M3
ES-MS2	Node Core in production: at least one representative service integrated with the EOSC Federated AAI through RedIRIS SIR and eduGAIN (FC-1), and at least one national catalogue or registry, RECOLECTA as first candidate, harvested into the EOSC Federation catalogue ecosystem (FC-2).	M6
ES-MS3	First onboarding wave and first training: initial portfolio discoverable and accessible through the Federation, including the resources of the four use cases with access policy and TRL; ES-UC3 federated workspace in operation for the Euclid DR1 task force; first training session delivered to the Node community.	M9
ES-MS4	Node operations and first use cases: FC-3 helpdesk and FC-4 monitoring in production; ES-UC3 and ES-UC4 demonstrated end to end (discovery, single authentication, analysis next to the data, citable FAIR research products); first annual EOSC-ES event held, third-party onboarding opened and at least 50% of the committed resources onboarded; mid-term self-assessment against the Federation minimum requirements presented to the technical committee.	M12
ES-MS5	Service management and governed access: FC-5 Service Management System established; ES-UC1 governed-access rehearsal on an already-permitted or synthetic cohort, with the privacy tooling in the loop and the cross-border pathway through HealthData@EU specified.	M15
ES-MS6	ES-UC1 and ES-UC2 demonstrated end to end with their counterpart Nodes (reported in ES-D4).	M18
ES-MS7	Node at full scope: all committed resources onboarded and second annual EOSC-ES event held (reported in ES-D5).	M24

Funded by the European Union
Grant agreement ID: 101188045

9 Contacts

The EOSC-ES Node is coordinated by the IACS, with a general assembly, a strategic committee where funding bodies are represented, and a technical committee responsible for the Federating Capabilities and the resource portfolio. The tables below give the five roles required by the Federation and the technical owner of each Federating Capability; the open entries follow the designation process described after them.

Role	Name	Email
Coordinator	[TBD] (IACS)	[TBD]
Operations Officer	[TBD]	[TBD]
Cybersecurity Officer	[TBD]	[TBD]
Legal Officer	[TBD]	[TBD]
Communications Officer	[TBD]	[TBD]
Federating Capability	**Technical owner**	**Institution**
FC-1 AAI	[TBD]	RedIRIS (Red.es)
FC-2 Catalogues and Registry	[TBD]	RedIRIS / FECYT
FC-3 Helpdesk	[TBD]	UPV / IACS / UJAEN
FC-4 Monitoring	[TBD]	[TBD]
FC-5 Service Management System	[TBD]	[TBD]

The number of individuals from the participating institutions for the duration of the build-up phase is still under consideration. Designating the representing entity and the remaining role holders is a governance decision rather than an administrative one: a Node federating 22 institutions must take it by agreement within the consortium and in a wider discussion with the national policy actors that will endorse and sustain it. It opens at the Spanish tripartite event in Zaragoza on 2 October 2026, with the EOSC Association, the European Commission, the EOSC Steering Board, COS-Gob and the Ministry of Science, Innovation and Universities, on an agenda that runs from this Charter to the enrolment application. Even though some of the responsible institutions have been identified, all roles and technical owners will be named ahead of ES-MS0.

Funded by the European Union
Grant agreement ID: 101188045